



**University of Sharjah**  
**College of Computing and Informatics**  
Department of Computer Science  
1501491: Topics in Computer Science II  
Final Project Report, Spring 25-26  
Due Date: Monday, April 20<sup>th</sup>, 2026

## **Handwritten Word Recognition and Writer Identification Using CNNs**

### **Group Members:**

Aisha Mohammed - [REDACTED]  
Ela Hammami - [REDACTED]  
Mariam Albannay - [REDACTED]  
Noorah Alshamsi - [REDACTED]  
Shaikha Alfalasi - [REDACTED]

## **Abstract**

This project presents a comprehensive deep learning framework for Handwritten Word Recognition and Writer Identification using Convolutional Neural Networks (CNNs). The system was trained on two datasets: the EMNIST Letters dataset and a personal dataset of handwritten words that were obtained by the five team members. The Word Recognition model (CharCNN) utilizes a three-layer convolutional architecture to extract visual features, combined with contour-based segmentation and dictionary-based correction, which achieved a character-level accuracy of 93.61% on the EMNIST test set and a final word-level accuracy of 66.67% on the custom dataset. The writer identification model (WriterCNN) labels handwriting samples of five familiar writers and works with both closed-set and open-set tasks with the accuracy of 93.33% and a macro F1-score of 0.9314. A prototype-based open-set recognition system with distance thresholds was also introduced that attained a 90% rejection rate on unknown writers. The final system was integrated with a web application, deployed on April 14, 2026, to show the applicability of the system in real-time analysis of handwriting.

# Table of Contents

|                                 |           |
|---------------------------------|-----------|
| <b>Section 1.....</b>           | <b>3</b>  |
| 1.1 Background.....             | 3         |
| 1.2 Problem Statement.....      | 3         |
| 1.3 Objectives.....             | 3         |
| <b>Section 2.....</b>           | <b>4</b>  |
| 2.1 Related Work.....           | 4         |
| <b>Section 3.....</b>           | <b>5</b>  |
| 3.1 Dataset.....                | 5         |
| 3.2 Data Preprocessing.....     | 6         |
| 3.3 CNN Architecture.....       | 7         |
| 3.4 Training Configuration..... | 8         |
| <b>Section 4.....</b>           | <b>9</b>  |
| 4.1 Training Results.....       | 9         |
| 4.2 Model Evaluation.....       | 10        |
| <b>Section 5.....</b>           | <b>13</b> |
| 5.1 References.....             | 13        |

# Section 1

## 1.1 Background

Handwriting Recognition refers to the conversion of handwritten text images to machine readable text. Old methods were based on manually crafted feature extraction methods which frequently were unable to generalise between different handwritings. As the field of deep learning has evolved, Convolutional Neural Networks (CNNs) have increasingly been a popular method to extract visual features like edges, strokes, and character shapes directly out of raw images, which have been shown to dramatically increase recognition accuracy. [1]

Writer Identification is the challenge of identifying an author of a written sample using unique writing attributes like letter shapes, writing spacing, and writing style. This issue is regarded as a type of biometric identification since every person has a distinct handwriting style. CNNs and other deep learning models have demonstrated high abilities in learning such discriminative features and differentiating among writers. [2]

## 1.2 Problem Statement

This project addresses the problem of handwritten word recognition, which is still a difficult problem to solve, as the styles of writing, the shapes of characters, their spacing, and quality of images all differ. Such changes may cause misclassification, particularly in case of small datasets. Even deep learning models, such as CNNs, might not be able to achieve consistent accuracy on different samples of handwriting, therefore reliable recognition of words in practice becomes hard.

In addition, the project also addresses the issue of writer identification based on a small dataset of handwritten samples of five people. One of the most important problems is the ability to differentiate between writers with similar handwriting, as well as to correctly reject samples of writers that are not in the training. Similar characteristics like stroke style, spacing, and structure complicate classification especially where data is scarce, which shows the need for more robust identification methods.

## 1.3 Objectives

The following are the objectives of this project:

- Develop a CNN-based handwritten word recognition system for accurate word prediction.
- Train a CNN-based writer identification model to classify samples among the five known writers.
- Implement an open-set recognition method to reject unknown writers using prototype embeddings.
- Build and preprocess a custom handwritten dataset collected from the five team members.
- Achieve strong performance in both word prediction and writer identification tasks.

## Section 2

### 2.1 Related Work

#### **Study 1: Handwritten Character Recognition Using Convolutional Neural Network**

The proposed study suggests the use of a handwritten character recognition system, which is built on Convolutional Neural Networks (CNNs), in which the input images are initially preprocessed and segmented to extract separate characters in words or text lines. The segmented characters are then resized and fed into the CNN which automatically derives features like edges, curves and shapes using convolutional and pooling layers unlike manually engineered features. These characteristics are then inputted into fully connected layers to categorize each character into the label that it represents. The NIST data is used to train and test the model, which proves that deep learning is effective in dealing with handwriting variations, with an accuracy of approximately 92.91%. [3]

#### **Study 2: Handwritten Character Recognition: A Comprehensive Review**

The paper provides a detailed methodology of handwritten character recognition through a multi-stage pipeline, which includes preprocessing, segmentation, feature extraction and classification. Input images are cleaned and normalized first, and the segmentation to isolate individual characters is done. Such features as edges and strokes are obtained, commonly with the assistance of Convolutional Neural Networks (CNNs). To classify, the techniques employed include Artificial Neural Networks (ANNs), Support Vector Machines (SVMs), and k-Nearest Neighbors (k-NN) which are used to classify each character by its label. Since it is a review paper, it is not based on one dataset but rather includes discussions on popular datasets like MNIST and EMNIST, pointing out that the current state of deep learning methods can attain high accuracy on each of those datasets. [4]

#### **Study 3: A Novel Methodology for Offline English Handwritten Character Recognition Using ELBP-Based Sequential CNN**

The paper is a hybrid suggestion of offline handwritten character recognition, which involves the combination of Enhanced Local Binary Pattern (ELBP) with a sequential Convolutional Neural Network (CNN). The input images are first preprocessed and made ready and then ELBP is used to isolate texture-based features in detail by use of extended pixel neighborhoods and logical operations like AND and OR. The resulting features are then inputted into a multi-layer CNN that is composed of convolutional, pooling, dropout and fully connected layers, which are automatically trained to learn deeper representations and classify them using a softmax layer. This model is trained and tested on the EMNIST dataset, showing that a combination of handcrafted ELBP features with CNN enhances recognition performance, with an accuracy of about 87.31%. [5]

#### **Study 4: End-To-End Evaluation of Deep Learning Architectures for Offline Handwriting Writer Identification: A Comparative Study**

The paper introduces an end-to-end system of offline handwriting writer identification that compares various deep networks in a single pipeline. Input images are initially processed by converting them to grayscale, eliminating noise, thresholding and resizing them to maintain a steady quality. This system then uses Convolutional Neural Networks (CNNs) to automatically create features of word-level handwritten images, without manual feature engineering. Various architectures such as VGG16, ResNet50, MobileNet, Xception and EfficientNetB0 are used to determine the writer using learned representations. The models are tested and trained on the IAM and CVL datasets and the results indicate that, although ResNet50 has the highest training accuracy, Xception has better generalization because the difference between training and validation performance is less. [6]

#### **Study 5: An Enhanced Hybrid Model Based on CNN and BiLSTM for Identifying Individuals via Handwriting Analysis**

This research suggests a hybrid method for handwriting identification involving the use of Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term memory (BiLSTM). The original images are initially preprocessed and analyzed to obtain various types of features, such as statistical, spatial and kinematic features and then the features are selected through methods like ANOVA and Mutual Information to eliminate the least significant information. The features of interest are then inputted into a CNN to detect visual patterns, and the BiLSTM component is used to learn sequential dependencies in the handwriting data. The joint architecture allows the representation and classification to be more effective than when a single model is used. The model is tested on a Bengali handwriting dataset, and it has shown that the hybrid CNN-BiLSTM model can perform better than the single models. [7]

## **Section 3**

### **3.1 Dataset**

The primary dataset used is the **EMNIST Letters dataset**, which has a total of 145,600 grayscale images of handwritten English letters from ‘a’ to ‘z’. The dataset comprises 124,800 training images and 20,800 testing images. The size of each image is fixed and equal to 28×28 pixels, and the data is saved in CSV format, with the first column being the label and the remaining 784 values being the pixels’ intensities. Each of the 26 classes in the dataset has 4,800 training images and 800 test images, thereby guaranteeing a fair representation of all the letters in the training process.

In addition to EMNIST, a separate **customized dataset** was created with handwritten word images collected from group members. The dataset is further split into two folders: training and testing. The training folder contains five subfolders, one folder for each group member (Aisha,

Ela, Mariam, Noorah, and Shaikha) and each folder contains 46 handwritten word images in .png format. These images follow the same format of “word\_wordNumber.png” (e.g., “grain\_2.png”). The testing folder contains 30 images in total, named using “img\_xx.png” format. This helps the model to be trained on different handwriting styles and to be tested on new data.

## 3.2 Data Preprocessing

### Model 1: Word Recognition

Before training, the EMNIST images are first reshaped to 28x28 format, oriented correctly using transposition, and normalized to the range [0, 1]. They are then scaled to tensors with a channel dimension to fit CNN input needs, while labels are scaled from 1-26 to 0-25. In the case of the custom dataset, word images are turned into grayscale, smoothed with the help of the Gaussian blur, and binarized with the help of the thresholding. Dilation is used to increase the visibility of characters, and contour detection is used to isolate characters. Noise is eliminated by filtering small or overlapping contours. The characters found are ordered in left-to-right order to preserve the order of letters, resized to keep the aspect ratio, padded to 28×28 pixels, normalized, and then transformed into tensors. Also, the gaps between bounding boxes are used to analyze the character spacing to identify the spaces between words and an adaptive threshold is calculated per image. This is then followed by a dictionary-based spelling correction step with the help of the **pyspellchecker** library that allows refining the predicted words by correcting any potential spelling errors to enhance the overall level of readability and accuracy.

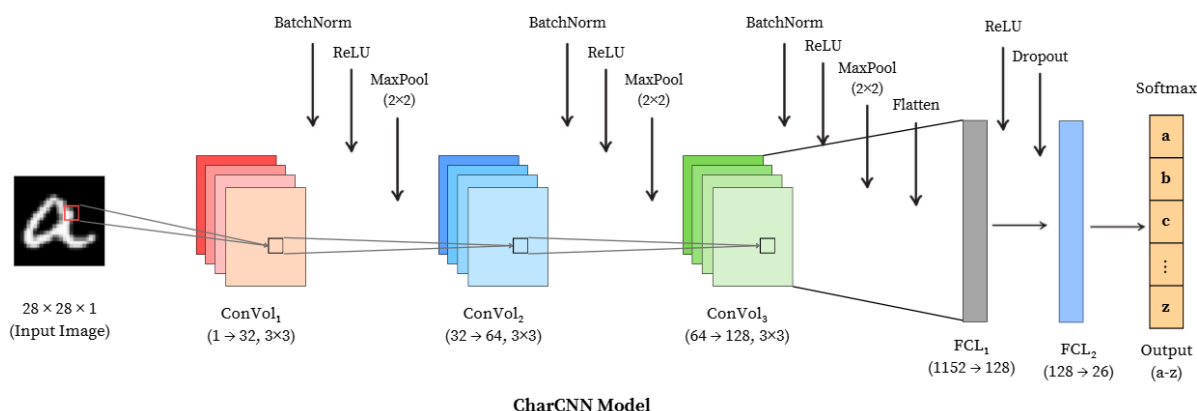
### Model 2: Writer Identification

The writer identification model preprocessing pipeline was implemented in the WriterWordDataset class, and it aimed at generating clean and standardised input tensors without losing stylistic features of each writer. Images were then turned to grayscale with the PIL "L" mode followed by a blank-margin trimming step where pixel rows and columns with minimum intensity below a threshold of 250 were identified and the relevant margins were trimmed without affecting the handwritten text. The image was then cropped and resized to a constant height of 64 pixels with aspect ratio maintained and scaled proportionally so that width was not more than 256 pixels. The resulting image was pasted on a white, 256×64 pixel canvas, vertical centering was done to provide a similar spatial localization of all samples. The pixel values were normalised to the range [0, 255] to [0.0, 1.0] and transformed into a single channel PyTorch-tensor containing the shape [1, 64, 256]. The writer-preserving optional data augmentation in training included small random rotations ( $\pm 4^\circ$ ), slight affine translations and scaling ( $\pm 4\%$  and 0.93-1.07) as well as low-magnitude Gaussian noise (sigma = 0.025 and applied with 60% probability), and minor brightness jitter (factor 0.9-1.1, and applied with 40% probability).

### 3.3 CNN Architecture

#### Model 1: Word Recognition

CharCNN is a convolutional neural network model for classifying single handwritten characters. It takes a  $28 \times 28$  greyscale image as input and sends it through three convolutional layers that identify and extract important visual features like edges, strokes, etc. Each convolutional block contains a convolution layer followed by batch normalization to stabilize training, ReLU activation to introduce non-linearity, and max pooling ( $2 \times 2$ ) to reduce spatial while keeping important features. The resulting feature maps are then flattened into a one dimensional vector to be used in the classification. This vector is then fed to the fully connected layers, where the learned features are combined to produce a final decision, with a dropout layer to help reduce overfitting. The last output layer consists of 26 neurons that correspond to the letters a-z allowing the model to classify each letter.



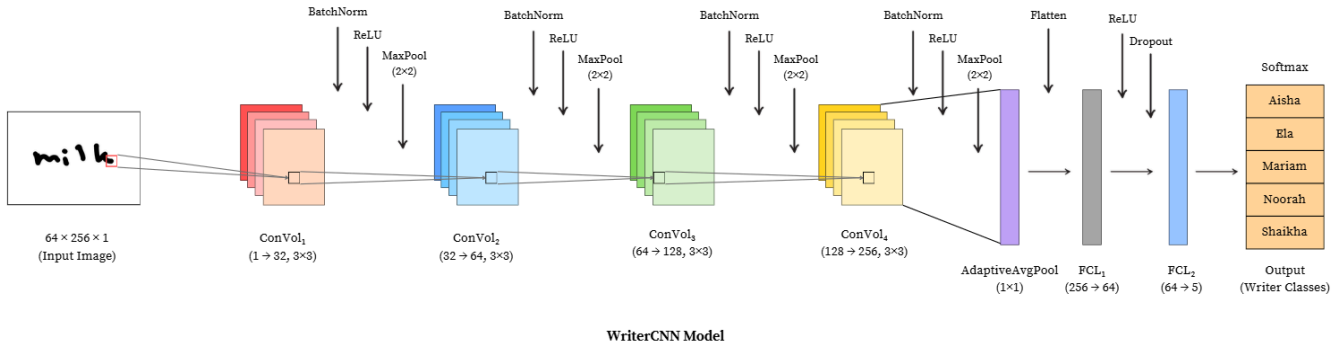
**Figure 1:** Illustration of the CharCNN model architecture

#### Model 2: Writer Identification

The model of writer identification is constructed in the form of a Convolutional Neural Network, which is referred to as WriterCNN. It accepts a grayscale word image of  $64 \times 256$  pixels as the input, and processes it through four convolutional blocks. The blocks have the same structure: a convolutional block that searches the image with visual patterns, a batch normalisation block to ensure the training is stable, a ReLU activation block to introduce non-linearity, and a max pooling block to reduce the spatial size by half. The size of the feature maps increases in the four blocks from 32 to 64, to 128, and finally to 256, which enables the model to identify more complex features of handwriting as it becomes deeper. Following the four blocks, an adaptive average pooling layer shrinks the entire feature map to a single 256-dimensional vector no matter what size the remaining spatial size may be. This is then



embedded into an embedding layer, which is a fully connected layer that further reduces it to a 64-dimensional representation, then activated by a ReLU, and a dropout layer with a dropout rate of 0.45 to minimize overfitting. That 64-dimensional output which the model considers as a small fingerprint of the writing style of the writer. Lastly, a linear classifier layer computes the fingerprint to a prediction between the five known writers. In open-set evaluation, the identical 64-dimensional embedding is directly extracted and compared with stored writer prototypes using Euclidean distance, rather than using the classifier output.



**Figure 2:** Illustration of the WriterCNN model architecture

### 3.4 Training Configuration

#### Model 1: Word Recognition

This model is trained with Adam optimizer with an initial learning rate of 0.001 to pretrain on EMNIST and a learning rate of 0.0005 on the fine-tuning on the custom dataset. Cross-entropy loss is used for classification, as the task is a multi-class problem with 26 output classes corresponding to the letters (a-z). The training is done in batches and a batch size of 64 is used in EMNIST and 32 in the custom dataset. The model is trained on the EMNIST dataset in 10 epochs, where it learns general character features, and then 3 epochs are done to fine-tune the model to the handwriting styles in the custom dataset. Fine-tuning is done with fewer epochs to decrease the chances of overfitting due to the small size of the dataset.

#### Model 2: Writer Identification

The AdamW optimizer was used to train the WriterCNN model with an initial learning rate of 0.0006 and a weight decay rate of 0.0002, helping in avoiding overfitting of the model through large weights. The cross-entropy loss was taken with a small label smoothing value of 0.05, a small value that smooths the target labels during training to ensure that the model is not overconfident in its predictions. The maximum number of epochs and a batch size of 16 were established to be used in training. A learning rate scheduler was applied to the model to

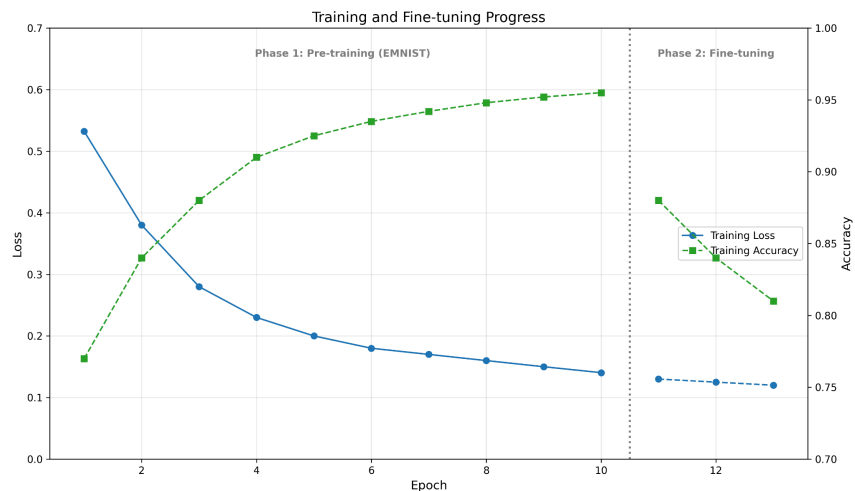
improve its learning over time, namely ReduceLROnPlateau, which tracks the validation macro F1 score and decreases the learning rate every three epochs without improvement. Early stopping was also used and the patience was set to 8 epochs, which implies that the training would terminate fully, unless the validation macro F1 score improved in 8 consecutive epochs. The most optimal model checkpoint was stored depending on the best validation macro F1 score obtained throughout training. In practice, training stopped at epoch 22 due to early stopping, with the best model found at epoch 14, achieving a perfect validation accuracy and macro F1 score of 1.0.

## Section 4

### 4.1 Training Results

#### Model 1: Word Recognition

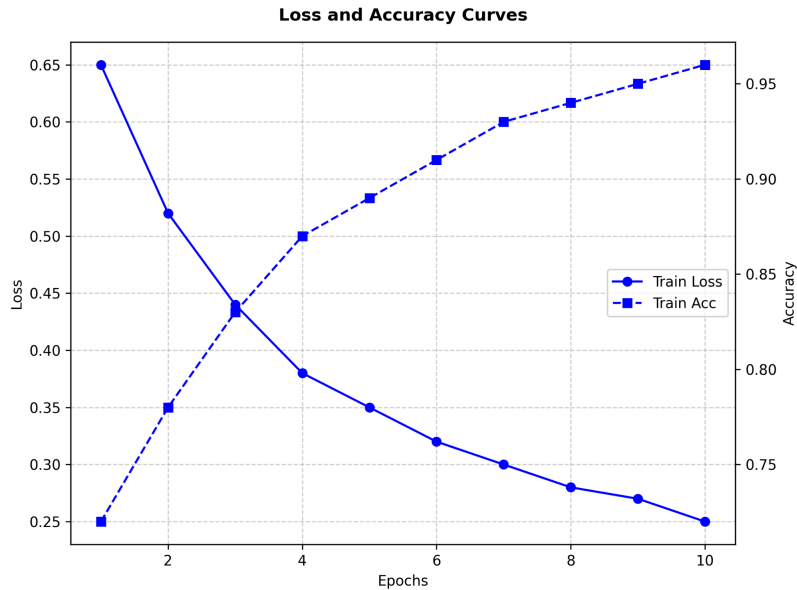
Recognition model training was done in two phases, and the measurement was made at both the word and character level. The training loss decreased gradually throughout all the epochs, with the initial training loss being 0.5321 during the first epoch and decreasing to 0.1404 in the tenth epoch, indicating that the model was learning over time. Subsequent to pretraining, the model was tested on the EMNIST test set, and its overall character level accuracy was 93.61%. During the second stage, weights were slowly adjusted to accommodate the custom handwriting styles without forgetting the previously learned features. Fine tuning was only done with samples where the number of segmented characters had the same length as the actual word length to obtain clean training data. The model was then fine-tuned and tested on a 30 image set of handwritten words. It scored a final word-level accuracy of 66.67% (20/30 words correctly predicted) and a character level accuracy of 79.69%, even when the entire word was not accurately predicted, most characters in that word were.



**Figure 3:** Illustration of the training loss and accuracy of the word recognition model

## Model 2: Writer Identification

The model of writer identification was tested on a test set of 30 samples, 6 images of each writer in the five known writers: Aisha, Ela, Mariam, Noorah, and Shaikha. Under the closed-set classification task, the model only predicts which of the five writers wrote a given word, the model achieved a test accuracy of 93.33% and a macro F1 of 0.9314, with a test loss of 0.3408. Under the per-writer results, three writers (Ela, Mariam and Noorah) were perfectly identified with a precision, recall, and F1 score of 1.0. The model correctly recalled 66.67 out of 100 of Aisha's samples, and thus, was confused by Shaikha, but Shaikha had a slightly lower precision of 0.75 because of the misclassifications. To evaluate it in the open-set case, that is, when the model is also required to determine whether a handwriting sample is met by a totally new unknown writer, a distance threshold was established at the 85th percentile of the Euclidean distances in the validation set, that provides a threshold value of about 4.01. The model tested on 10 samples of an unknown writer who had not been observed during training recognized 9 of 10 samples as none of the group, with an unknown rejection rate of 90, indicating the prototype-based distance method is a fairly good approach at recognizing writers that are not in the training group.



**Figure 4:** Illustration of the training loss and accuracy of the writer identification model

## 4.2 Model Evaluation

### Model 1: Word Recognition

The recognition model was tested at two levels of measurement; character level accuracy and word level accuracy. On a character level, individual predicted letters were compared to the

individual true letters, providing a fine-grained picture of the level of model reading of individual characters in a word. At the word level, a prediction could only be considered correct at the word level when the whole word was an exact match of the actual label. The evaluation pipeline was the following: each test image was first processed by the predict\_word function, which applied contour-based segmentation to each image resulting in the individual characters and then subjected each to the trained CharCNN model to obtain a predicted letter and a confidence score. The confidence score was achieved by using the softmax on the raw output of the model, and the most probable class was selected as the predicted character. Gaps between segmented characters were analysed to detect spaces between words, using a dynamic threshold based on the median gap size and average character width. After assembling the raw word, it was inputted through the smart\_correct\_word function which utilized the difflib of Python to locate close dictionary matches by comparing the top 50,000 most commonly used English words, scored each candidate based on character-level confidence alignment, and word frequency (using Zipf scores), and only substituted the raw prediction with a better-scoring candidate with a score over 1. There was also a safety rule that guarded high-confidence correct predictions - in case the mean confidence was 90% or greater and the raw prediction was already a legal dictionary word, then no correction was made. The last assessment was executed on 30 test word images where both the number of words correctly predicted and the number of characters in the correct words were compared.

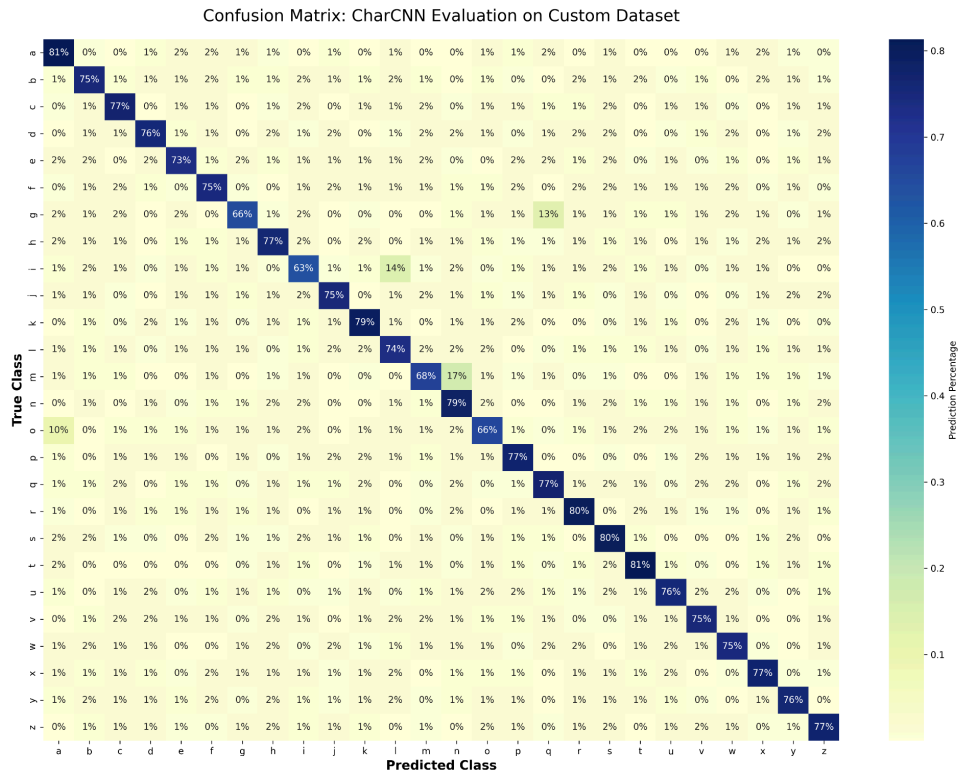
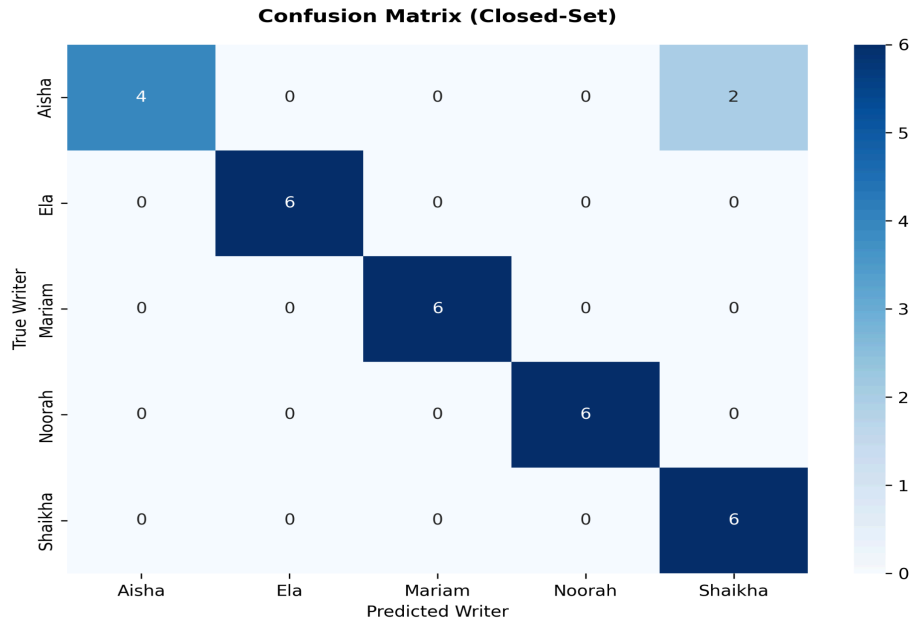


Figure 5: Confusion matrix of the CharCNN model

## Model 2: Writer Identification

Two different strategies were employed to assess the writer identification model in both closed-set and open-set cases. To evaluate the closed-set in the case of the closed-set evaluation, where the model would merely have to determine which of the five known writers wrote a given word, standard classification measures were employed: aggregate accuracy, macro-average F1-score, per-class precision and recall, and a confusion matrix created with scikit-learn. Macro F1 score was selected as the main measure to be used in the training since it considers all the five writers equally, irrespective of the sample size that each writer holds. In the open-set case, where the model also has to make a decision whether a sample is due to an entirely new writer that the model has never encountered before, a different method was necessary as standard classification measures cannot quantify this. The 64-dimensional embedding of the model was obtained on each test image and compared with per-writer prototype vectors - a prototype of each writer being the average embedding of all training samples of that writer - by use of Euclidean distance. Only those samples accepted as known writers had a distance to the nearest prototype that was below a fixed threshold and its softmax confidence score was at least 0.5. These two conditions were to be met simultaneously; otherwise, the sample was considered to be none of the group. The distance threshold itself was set by the validation set to the 85th percentile of the known-writer distances, which was about 4.01. The open-set performance was then quantified based on two rates, the known accept rate (the proportion of times known writers were correctly accepted), and the unknown rejection rate (the proportion of times the unknown writer was correctly rejected), and the final fused rule had a known accept rate of 83.33% and an unknown rejection rate of 90%.



**Figure 6:** Confusion matrix of the WriterCNN model

## Section 5

### 5.1 References

Handwriting Recognition: Live Web Application, Deployed April 14, 2026.

<https://handwriting-recognition-ehq7.onrender.com/>

[1] P. Baheti, "Handwriting Recognition: Definition, Techniques & Uses," V7 Labs, Dec. 2, 2022.

<https://www.v7labs.com/blog/handwriting-recognition-guide>

[2] V. Christlein, D. Bernecker, A. Maier, and E. Angelopoulou, "Offline writer identification using convolutional neural network activation features," *arXiv preprint arXiv:2402.17029*, 2024. Available:

<https://arxiv.org/abs/2402.17029>

[3] A. K. Singh, S. Sarkar, and D. Gupta, "Handwritten character recognition using convolutional neural network," *Journal of Physics: Conference Series*, vol. 1918, no. 4, 2021. Available:

<https://iopscience.iop.org/article/10.1088/1742-6596/1918/4/042152>

[4] M. A. Al-Maqaleh, A. M. Al-Hagery, and A. M. Al-Maqtari, "Handwritten Character Recognition: A Comprehensive Review," *Central Asian Journal of Mathematical Theory and Computer Sciences*, vol. 4, no. 3, 2023. Available: <https://cajmtcs.casjournal.org/index.php/CAJMTCS/article/view/568/608>

[5] M. Humayun, R. Siddiqi, M. Uddin, I. A. Kandhro, M. Abdelhaq, and R. Alsaqour, "A novel methodology for offline English handwritten character recognition using ELBP-based sequential CNN," *Neural Computing and Applications*, vol. 36, pp. 19139–19156, 2024. Available:

<https://doi.org/10.1007/s00521-024-10206-1>

[6] W. Suteddy, D. A. R. Agustini, A. Adiwilaga, and D. A. Atmanto, "End-to-end evaluation of deep learning architectures for offline handwriting writer identification: A comparative study," *International Journal on Informatics Visualization*, vol. 7, no. 1, pp. 178–185, 2023. Available:

<https://www.joiv.org/index.php/joiv/article/view/1293>

[7] M. A. H. Akhand, M. R. Islam, M. A. Hossain, and M. M. Rahman, "An enhanced hybrid model based on CNN and BiLSTM for identifying individuals via handwriting analysis," *Computer Modeling in Engineering & Sciences*, vol. 140, no. 2, pp. 1685–1705, 2024. Available:

<https://www.techscience.com/CMES/v140n2/56554>

| <b>Student declaration:</b>                                                                                                                                                                                                                                                                                                                                               |                  |                                                                                     |                                                       |         |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------|---------|-------|
| <p><b><i>I declare that:</i></b></p> <ul style="list-style-type: none"> <li>▪ <i>I understand what is meant by plagiarism</i></li> <li>▪ <i>The implications of plagiarism have been explained to us by our lecturer</i></li> <li>▪ <i>This project is all my work, and I have acknowledged any use of the published or unpublished works of other people.</i></li> </ul> |                  |                                                                                     |                                                       |         |       |
| <b>Group:</b>                                                                                                                                                                                                                                                                                                                                                             |                  |                                                                                     |                                                       |         |       |
| <b>Names of Candidate</b>                                                                                                                                                                                                                                                                                                                                                 |                  |                                                                                     |                                                       |         |       |
| No.                                                                                                                                                                                                                                                                                                                                                                       | Student Name     | Student ID                                                                          | Contribution                                          | Overall | Score |
| 1                                                                                                                                                                                                                                                                                                                                                                         | Aisha Mohammed   |    | - Section 3<br>- Writer identification model          |         |       |
| 2                                                                                                                                                                                                                                                                                                                                                                         | Ela Hammami      |  | - Section 4<br>- Deployment                           |         |       |
| 3                                                                                                                                                                                                                                                                                                                                                                         | Mariam Albannay  |  | - Section 1 and Section 2                             |         |       |
| 4                                                                                                                                                                                                                                                                                                                                                                         | Noorah Alshamsi  |  | - Section 1 and Section 2                             |         |       |
| 5                                                                                                                                                                                                                                                                                                                                                                         | Shaikha Alfalasi |  | - Section 2 and Section 3<br>- Word recognition model |         |       |